2DSegFormer: 2-D Transformer Model for Semantic Segmentation on Aerial Images

Xinyu Li, Yu Cheng, *Member, IEEE*, Yi Fang, *Senior Member, IEEE*, Hongmei Liang, and Shaoqiu Xu

***Abstract*— Two-dimensional position information of input tokens is essential for transformer-based semantic segmentation models, especially on high-resolution aerial images. However, recent transformer-based segmentation methods use position encoding to record position information and most position encoding methods encode the 1-D positions of tokens. Therefore, we propose a 2-D semantic transformer model (2DSegFormer) for semantic segmentation on aerial images. In 2DSegFormer, we design a novel 2-D positional attention to accurately record the 2-D position information required by the transformer. Furthermore, we design the dilated residual connection and use it instead of skip connection in the deep stages to get a larger receptive field. Skip connections are used in the shallow stages of 2DSegFormer to pass the details to the corresponding stages in the decoder. Experimental results on UAVid, Vaihingen, and AeroScapes datasets demonstrate the effectiveness of 2DSegFormer. Compared with the state-of-the-art methods, 2DSegFormer shows better performance and great robustness on three different datasets. In particular, 2DSegFormer-B2 achieves first place in the public ranking on the UAVid test set.**

***Index Terms*— 2-D positional attention, aerial images, semantic segmentation, transformer.**

I. Introduction

DEEP-learning algorithms promote rapid development in the field of remote sensing [1], [2], and semantic segmentation is one of the important directions [3], [4], [5], [6], [7], [8], [9], [10], [11]. Semantic segmentation on high-resolution aerial images has always been a hot research topic in the field of remote sensing [3], [4] and traffic monitoring [5]. Semantic segmentation is achieved by labeling each pixel with the class of its enclosing object or region [6] and plays an important role in practical applications such as vehicle detection [7], land cover mapping [8], urban planning [9], and road extraction [10], [11].

Manuscript received 3 September 2022; revised 22 October 2022; accepted 15 November 2022. Date of publication 21 November 2022; date of current version 1 December 2022. This work was supported in part by the National Science Foundation (NSF) of China under Grant 62071131, in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2022B1515020086, in part by the International Collaborative Research Program of Guangdong Science and Technology Department under Grant 2022A0505050070, and in part by the Industrial Research and Development Project of Haoyang Electronic Company Ltd., under Grant 2022440002001494. *(Corresponding author: Yu Cheng.)*

The authors are with the School of Information Engineering, Guangdong University of Technology, Guangzhou 510006, China (e-mail: lixiny98@163.com; chengyu@gdut.edu.cn; fangyi@gdut.edu.cn; cc201216@163.com; shaoqiuxu@gdut.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3223416

After the invention of the fully convolutional network (FCN) [6], the convolutional neural network (CNN) almost dominates the field of computer vision (CV). More recently, U-Net [12] and SegNet [13] that adopts the CNN-based encoder–decoder structure became a preferred framework for semantic segmentation. However, the locality of convolution makes CNN-based methods difficult to model global semantic interactions. Although dilated convolution [14] and attention mechanism are used to alleviate this problem, these methods do not directly encode the global semantics. In contrast, the success of the transformer in natural language processing (NLP) is due to its powerful ability to model global semantics interactions. Dosovitskiy et al. [15] first introduced the transformer into image classification and proposed the vision transformer (ViT). Subsequently, Zheng et al. [16] used ViT as the backbone to propose SEgmentation TRansformer (STER) and achieved competitive performance compared to CNN-based models on semantic segmentation.

When using the transformer as the backbone, the input image needs to be cropped at the same size and then these tokens (image patches) are flattened into a 1-D sequence (see Fig. 1). Since self-attention cannot capture the order of input tokens, position encoding is often used to record the order of tokens for the transformer. The current position encoding methods are mainly divided into absolute position encoding (APE) [17], [18] and relative position encoding (RPE) [19]. In particular, APE is obtained by encoding the absolute positions of the input tokens (1 to the maximum length of the sequence) and then the resulting position encoding vectors are added to the image embeddings. RPE is obtained by encoding the relative positions between input tokens so that it can help the transformer-based model retain the spatial relationships between input tokens better. Therefore, RPE is a better choice for dense prediction tasks such as semantic segmentation.

In ViT, SETR, and most segmentation methods based on ViTs, APE designed for NLP tasks that records the 1-D positions is directly used. However, 1-D positions cannot accurately reflect the relative positional relationships of tokens in 2-D space. As can be seen in Fig. 1, we use the same color to mark these tokens that are equidistant from the central black token. For example, the relative distances between all green tokens and the black token are equal in 2-D space. These relative distances are no longer equal to one another in the 1-D sequence of tokens. Therefore, it is necessary to develop more efficient methods to record the 2-D positions for ViTs.

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

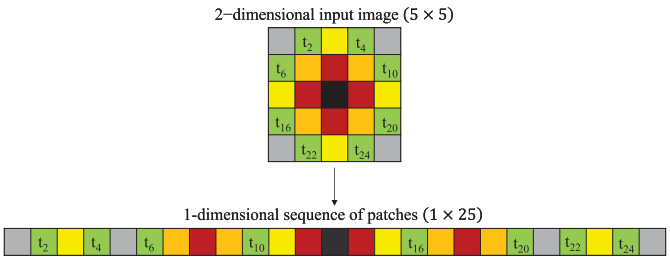


Fig. 1. Conversion process from the 2-D image to the 1-D sequence of tokens, where t_i represents the ith token from the left in the 1-D sequence. We use the black token as the reference token and use different colors to distinguish the distances from other tokens to the black token in the 2-D space. For example, the Euclidean distances between all green tokens and the central black token are $\sqrt{5}$ in 2-D space. However, in the 1-D sequence of tokens, the distances are 11, 9, 7, and 3, respectively.

Aerial images are taken by drones or high-altitude aircraft. Compared with semantic segmentation on general images, 2-D position information is more important for semantic segmentation on aerial images. This is because the shooting angle of the aerial images looking down causes a large number of objects to be relatively small in size and deformed. Utilizing 2-D position information in aerial images will help improve the performance of semantic segmentation.

Recently, some RPE methods [20], [21] that record the 2-D positions are proposed for ViTs. Compared with 1-D position encoding, 2-D position encoding enables ViTs to obtain more accurate 2-D position information and capture richer spatial details. In the work of Wu et al. [21], four new RPE methods based on 2-D positions are proposed. A many-to-one piecewise function is used to map relative distances to the indices of buckets filled with position encodings. Although this effectively reduces the number of parameters and computation costs, the accuracy of relative distances decreases after mapping. When the inputs are high-resolution aerial images, this method will bring a large amount of GPU memory costs and greatly reduce the inference speed.

In this article, we propose a 2-D semantic transformer model (2DSegFormer) for semantic segmentation on aerial images. A 2-D positional attention is conceived in the proposed 2DSegFormer to retain 2-D relative position information for ViTs. There are three major differences between our proposed 2-D positional attention and the existing RPE methods. First, we use attention weights, instead of vectors of learnable parameters, to record position information. This avoids the use of buckets and decreases the number of parameters for the encoder. Second, we use a continuous Gaussian function, instead of the many-to-one piecewise function, to map relative distances to attention values. This can effectively reduce the error caused by the mapping process to more accurately reflect the positional relationships. Third, the effective range of the proposed 2-D positional attention can be adjusted by varying the standard deviation of the Gaussian function. Thus, when compared with the existing RPE methods, the proposed 2-D positional attention is a more flexible method that fits better with the characteristics of the hierarchical feature maps extracted by the multistage encoder. The 2-D positional attention and self-attention are computed in parallel and the final self-attention weight is the weighted sum of the two. In addition, to capture more global context information in the deep stages, we design the dilated residual connection by using two dilated convolutional layers and the residual connection [22] to enlarge the receptive field of deep feature maps and preserve the original features, respectively. Our main contributions to this article can be summarized as follows.

1) We design a 2-D positional attention mechanism to retain the 2-D positional relationships between tokens, then integrate it into the self-attention mechanism. The final self-attention result is a weighted sum of self-attention and 2-D positional attention. Moreover, we set a learnable parameter as the weight of 2-D positional attention to let the model learn the weight of the two by itself.
2) We design the dilated residual connection and replace some skip connections with it to enlarge the receptive field. In this way, more semantic information and spatial details can be passed to the decoder.
3) Experimental results on different aerial image datasets show that 2DSegFormer can achieve fine segmentation and great robustness. Specifically, 2DSegFormer achieves first place in the public ranking on the official UAVid evaluation server and also shows excellent performance on the AeroScapes and Vaihingen datasets.

II. Related Work

A. CNN-Based Aerial Image Segmentation

After the invention of FCN [6] and U-Net [12], the CNN-based encoder–decoder structure is the preferred network structure. Zhao and Du [23] proposed a multiscale CNN to learn the spatial relationship between deep feature maps. Mou et al. [24] proposed the spatial relation module and the channel relation module to learn and reason about global relationships between spatial locations or feature maps, respectively. Wang et al. [25] proposed SaNet to improve the segmentation accuracy of scale-invariant objects by capturing multiscale context.

Limited by the locality of CNNs, using CNNs as the feature extractor results in a small effective receptive field and insufficient understanding of the global context. Therefore, many studies try to use dilated convolution, attention mechanism, or other methods to capture the global context better. Li et al. [26] proposed a self-smoothing atrous convolution to obtain a large and effective receptive field, which avoided gridding artifacts caused by the introduction of atrous convolution. He et al. [27] used a stack of global context attention modules to improve the global context representation capability of the model. Peng et al. [3] proposed CF-Net to improve the segmentation accuracy of small-size objects, which adopts channel attention to extract multiscale semantic information. Li et al. [28] proposed an attentive bilateral contextual network, which uses a spatial path and a contextual path to obtain fine spatial details and sufficient context information, respectively. Li et al. [29] proposed MANet that contains multiple efficient attention modules to extract context

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

dependencies. Niu et al. [30] proposed a novel attention-based framework to adaptively capture global correlations from the perspective of space, channel, and category. Chen et al. [31] added a self-attention module in the encoder to learn the attention weights of the input, thus enabling the network to pay more attention to positions where there may be salient regions. Besides, Huang et al. [32] proposed a multilevel context refinement network that adaptively aggregates context information at three levels, that is, local-level, image-level, and semantic-level. He et al. [33] proposed a semantic information-modulated deep subpixel mapping network to enhance the representation of spatial contextual features.

B. Transformer-Based Aerial Image Segmentation

Recently, the success of transformer in the field of CV due to its strong global context capture capability. After the invention of SETR, a transformer instead of a CNN is chosen as the feature extractor in many aerial image segmentation methods. Wang et al. [34] chose Swin Transformer [35] as the backbone and designed a densely connected feature aggregation module as the decoder. He et al. [36] proposed ST-UNet, which exploits the Swin transformer and CNN to form a dual encoder. Zhang et al. [37] used the Swin transformer as the backbone to extract features and used an atrous spatial pyramid pooling block to obtain a multiscale context. Wang et al. [38] proposed a bilateral awareness network that consists of a transformer-based dependency path and a CNN-based texture path to capture long-range relationships and fine-grained details in aerial images, respectively. Wang et al. [39] proposed a transformer-based decoder and developed an efficient global–local attention mechanism for modeling the global and local information of the decoder. Compared to CNN-based methods, the above transformer-based methods achieved competitive performance.

C. Position Encoding in Transformer

Fixed-length encodings generated by sine and cosine functions and the learnable encodings composed by training parameters [17], [18] are two common choices for APE. Shaw et al. [19] proposed RPE by modeling input tokens as a fully connected graph, where each edge in the graph is presented by a learnable vector that encodes the relative position between the two tokens connected by the edge.

The above position encoding methods are all designed for NLP tasks, considering the 1-D positions of input tokens. For CV tasks where the inputs are 2-D images, encoding 1-D positions cannot accurately reflect the relative positional relationships in 2-D space. Ramachandran et al. [20] decomposed 2-D relative distances into row offsets and column offsets and then concatenated the encodings of the two directions into a 2-D position encoding. In this way, the same encoding vector can be shared to reduce the number of parameters when the same offsets on a row or column. Wu et al. [21] proposed four RPE methods designed for 2-D images, called image RPE (iRPE). They designed bias and contextual modes to integrate iRPE into self-attention and proposed undirected and directed methods to study the importance of directivity. iRPE can achieve excellent performance on both image classification and object detection tasks. However, each encoding vector used in position encoding methods is a set of learnable parameters, which brings a certain number of parameters. Especially when the inputs are long sequences (e.g., high-resolution aerial images), this method will bring a large amount of GPU memory costs and greatly reduce the inference speed.

III. Method

A. Network Structure

2DSegFormer is based on the encoder–decoder structure and its overall architecture of it is shown in Fig. 2. In the encoder, we design a 2-D positional attention mechanism to retain the 2-D position information between tokens and integrate it into the efficient self-attention of Mix Transformer (MiT) [40]. The proposed 2-D positional attention mechanism provides richer spatial details required for semantic segmentation on aerial images to segment small-size objects. The encoder is divided into four stages similar to the design principles of ResNet [22]. Specifically, each stage contains two transformer layers and an overlapping patch merging module to extract features and down-sample the feature maps, respectively. Therefore, multilevel feature maps at {1/4, 1/8, 1/16, 1/32} of the original image resolution can be obtained. In the connection part of the encoder and decoder, we design the dilated residual connection to replace skip connections in the deep stages. The proposed dilated residual connection is used to capture richer global semantic context information, which can help the model distinguish confusing objects from different classes in aerial images better. Also, skip connections are used in the shallow stages to pass the details to the decoder. Finally, a stepwise up-sampling decoder is used to finely segment the extracted feature maps.

B. MiT Block

The standard transformer layer consists of a multihead self-attention mechanism (MSA) and a fully connected feed-forward network (FFN). Therefore, the output of the lth transformer layer can be expressed as

$$\boldsymbol{S}_l = \mathrm{MSA}(\boldsymbol{S}_{l-1}) + \mathrm{FFN}(\mathrm{MSA}(\boldsymbol{S}_{l-1})) \in \mathbb{R}^{N\times C} \tag{1}$$

where N and C represent the sequence length and the number of channels of S_l, respectively. Before being fed into transformer layers, input tokens or feature sequences are linearly projected to get the query (Q), key (K), and value (V) with the same dimensions $N \times C$. The calculation process of MSA can be expressed as follows:

$$\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} = \boldsymbol{S}_{l-1}\boldsymbol{W}_Q, \boldsymbol{S}_{l-1}\boldsymbol{W}_K, \boldsymbol{S}_{l-1}\boldsymbol{W}_V \tag{2}$$

$$\mathrm{MSA}(\boldsymbol{S}_{l-1}) = \boldsymbol{S}_{l-1} + \mathrm{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_{\mathrm{head}}}}\right)\boldsymbol{V} \tag{3}$$

where d_{head} is a scaling factor. The computational complexity of this process is $O(N^2)$. For a high-resolution image, it will cause a huge amount of calculation. Therefore, the efficient self-attention [40] in MiT is proposed to reduce the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

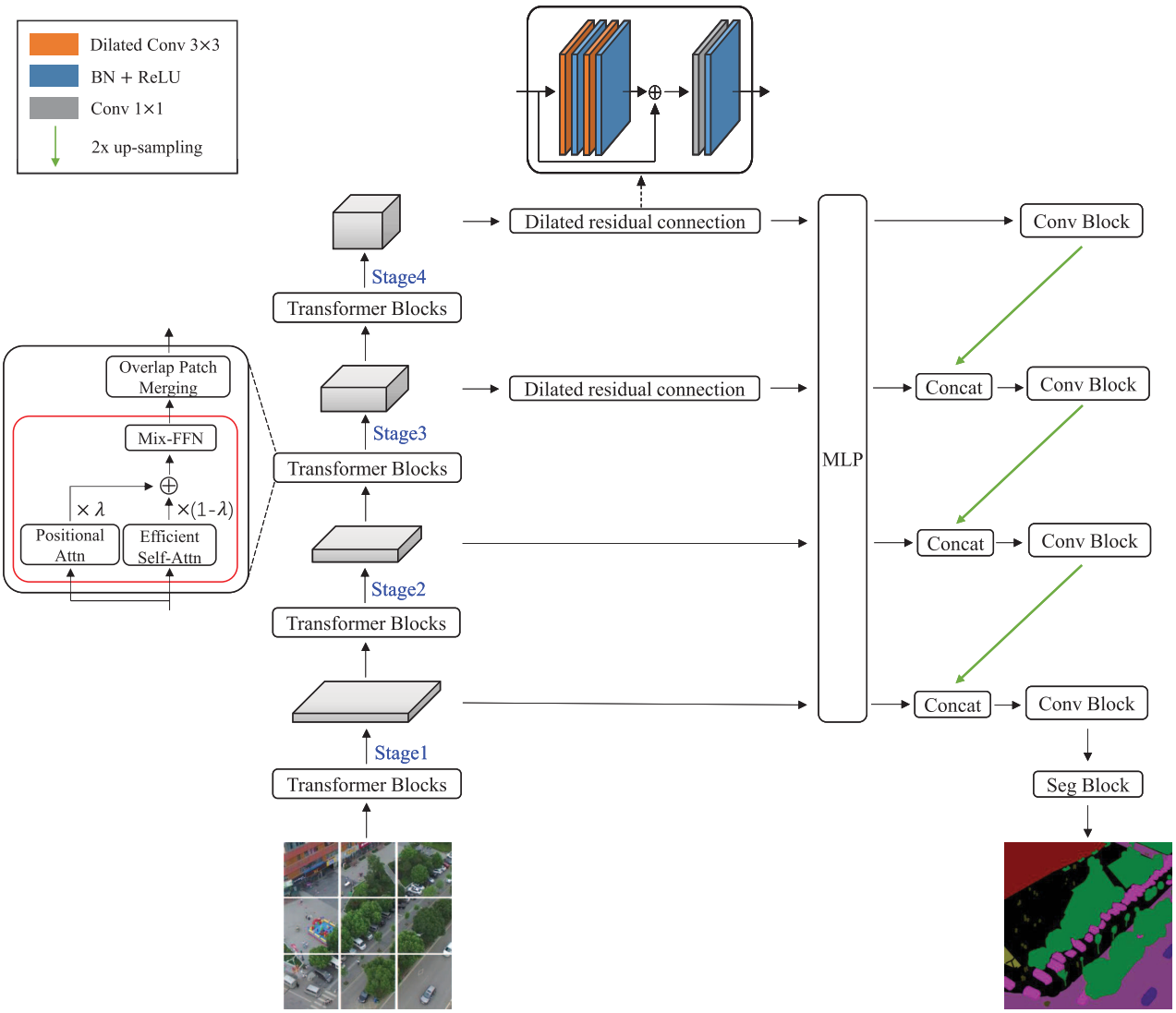


Fig. 2. Overall architecture of 2DSegFormer. It consists of a transformer-based encoder, dilated residual connections, and a stepwise up-sampling decoder. The internal structures of transformer blocks and the dilated residual connection are marked by dotted lines.

computational complexity by reducing the sequence length of sequences of $\boldsymbol{K}$ and $\boldsymbol{V}$. The method formulation is given as follows:

$$\boldsymbol{K}^{(N\times C)}, \boldsymbol{V}^{(N\times C)} \xrightarrow{\text{Reshape}} \boldsymbol{K}'^{\left(\frac{N}{R^2}\times C\cdot R^2\right)}, \boldsymbol{V}'^{\left(\frac{N}{R^2}\times C\cdot R^2\right)} \tag{4}$$

$$\boldsymbol{K}, \boldsymbol{V} = \text{Linear}\left(C\cdot R^2, C\right)\left(\boldsymbol{K}', \boldsymbol{V}'\right) \tag{5}$$

where Linear $(C_{\text{in}}, C_{\text{out}})(\cdot)$ refers to a linear layer taking a C_{in}-dimensional tensor as input and generating a C_{out}-dimensional tensor as output and R is the reduction ratio. As a result, the dimension of $\boldsymbol{QK}^{\top}$ in (3) is $N \times (N/R^2)$ and the computational complexity is reduced from $O(N^2)$ to $O(N^2/R^2)$. The calculation process of efficient self-attention is shown in Fig. 3(a).

C. 2-D Positional Attention

To design a transformer-based model for semantic segmentation on aerial images, we consider the 2-D positional relationships between input tokens and propose a 2-D positional attention mechanism to retain the 2-D relative position information. Different from the commonly used position encoding methods, we use the attention mechanism to measure the positional importance between different tokens. The calculation process of our self-attention with 2-D positional attention is shown in Fig. 3(b) and the 2-D positional attention is defined below.

Definition 1: The 2-D positional attention mechanism is a function 2-D_At$(\text{Lt}(t_i), \text{Lt}(t_j))$ that calculates how much attention the token t_i should pay to other tokens according to their 2-D distances. 2-D_At$(\text{Lt}(t_i), \text{Lt}(t_j))$ satisfy the following two properties.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

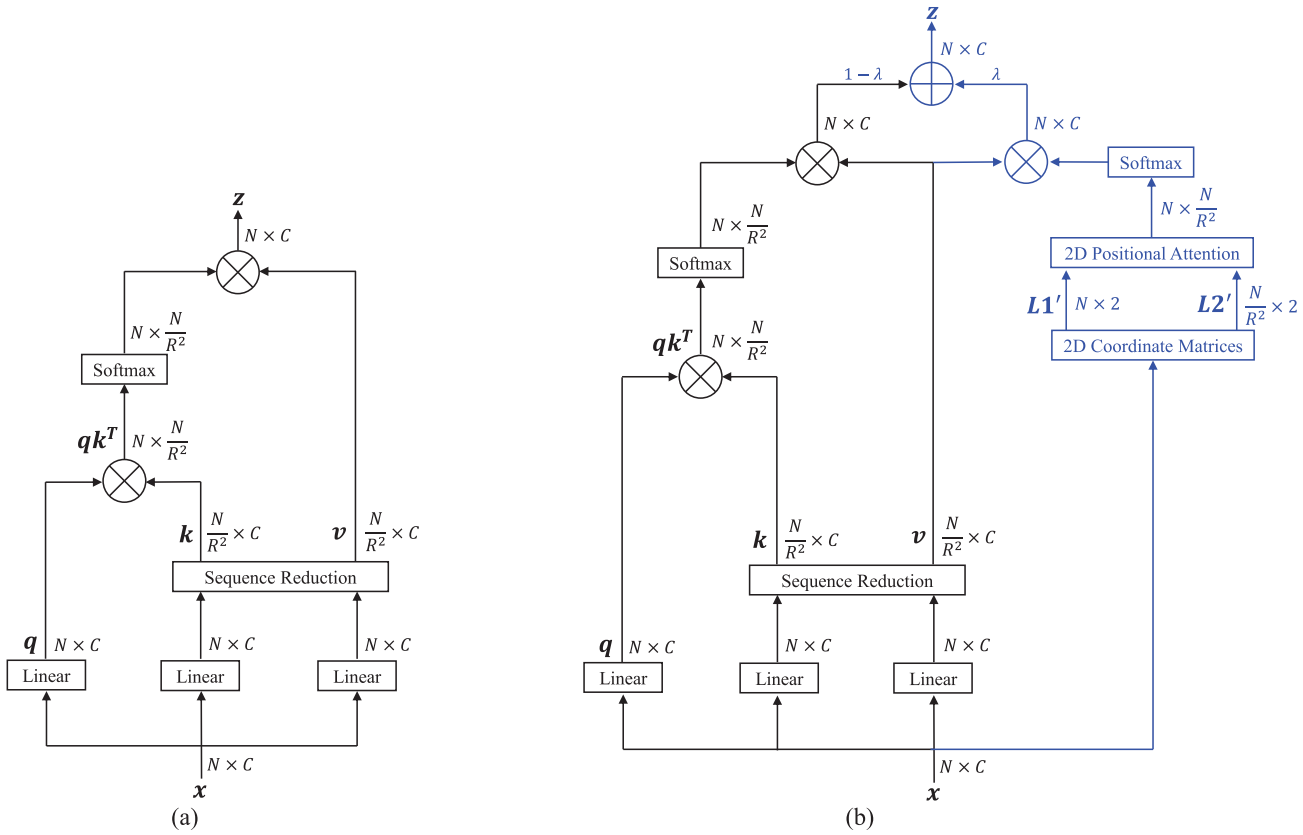


Fig. 3. Illustration of efficient self-attention and our self-attention with 2-D positional attention in 2DSegFormer. (a) Efficient self-attention proposed in the work of Xie et al. [40] and (b) integration of efficient self-attention and 2-D positional attention in 2DSegFormer. The blue parts are newly added to compute 2-D positional attention from spatial coordinate matrices $L1'$ and $L2'$.

1) 2-D_At(Lt(t_i), Lt(t_j)) ≥ 0 , where Lt(t_i) = [(x_i, y_i)] is the spatial coordinates of token t_i.
2) 2-D_At(Lt(t_i), Lt(t_j)) $\geq$ 2-D_At(Lt(t_i), Lt(t_k)), if 2-D_Dt(Lt(t_i), Lt(t_j)) $\leq$ 2-D_Dt(Lt(t_i), Lt(t_k)), where 2-D_Dt($\cdot$, $\cdot$) is the 2-D distance function.

For each token t_i, the 2-D positional attention is a vector $pa_i = [pa_i^{t_1}, \ldots, pa_i^{t_N}]$, where $pa_i^{t_j}$ = 2-D_At(Lt(t_i), Lt(t_j)) is the 2-D positional attention which t_i pays to t_j, and $j = 1, \ldots, N$, N is the number of tokens. The semantic attention of t_i is calculated by the transformer model. The semantic attention of token t_i is $sa_i = [sa_i^{t_1}, \ldots, sa_i^{t_N}]$, where $sa_i^{t_j}$ is the semantic attention which t_i pay to t_j, and $j = 1, \ldots, N$. The final self-attention sfa_i is the weighted sum of 2-D positional attention pa_i and semantic-attention sa_i as follows:

$$sfa_i = (1 - \lambda) * sa_i + \lambda * pa_i \tag{6}$$

where λ is a weight to adjust the proportion between self-attention and 2-D positional attention. In the experiments, we set λ as a trainable parameter.

Specifically, we reshape $\boldsymbol{S}$, which is the input of the transformer block with dimension $N \times C$, to a 2-D feature map $\boldsymbol{M}$ with dimension $\sqrt{N} \times \sqrt{N} \times C$ as follows:

$$\boldsymbol{S}^{(N\times C)} \xrightarrow{\text{Reshape}} \boldsymbol{M}^{(\sqrt{N}\times\sqrt{N}\times C)}. \tag{7}$$

This reshaping process is shown in Fig. 4. Then we obtain the spatial coordinates of the tokens in the 2-D space.

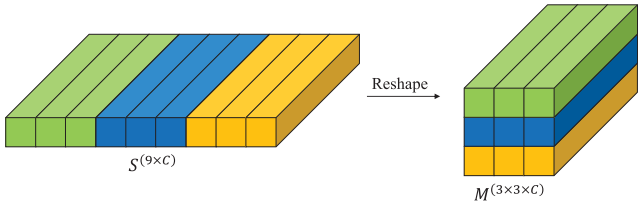


Fig. 4. Process of reshaping the 1-D sequence input of the transformer block $\boldsymbol{S}^{(N\times C)}$ to the 2-D feature map $\boldsymbol{M}^{(\sqrt{N}\times\sqrt{N}\times C)}$, where N is set to 9.

As shown in (8), we get the matrix $\boldsymbol{L1}$ containing the coordinates of $\sqrt{N} \times \sqrt{N}$ tokens in 2-D feature map $\boldsymbol{M}$. The value of an element $\boldsymbol{L1}[i, j]$ is the corresponding 2-D coordinate of the token

$$\boldsymbol{L1} = \text{Lt}(\boldsymbol{M}) = \begin{bmatrix} (0,0) & \cdots & \left(0, \sqrt{N}-1\right) \\ \vdots & \ddots & \vdots \\ \left(\sqrt{N}-1, 0\right) & \cdots & \left(\sqrt{N}-1, \sqrt{N}-1\right) \end{bmatrix}. \tag{8}$$

In the shallow stages of the encoder, the number of tokens is large. Therefore, we are inspired by the efficient self-attention in MiT and design a convolution method to reduce the computational complexity of 2-D positional attention. Instead of computing the distances between all tokens, we use R

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

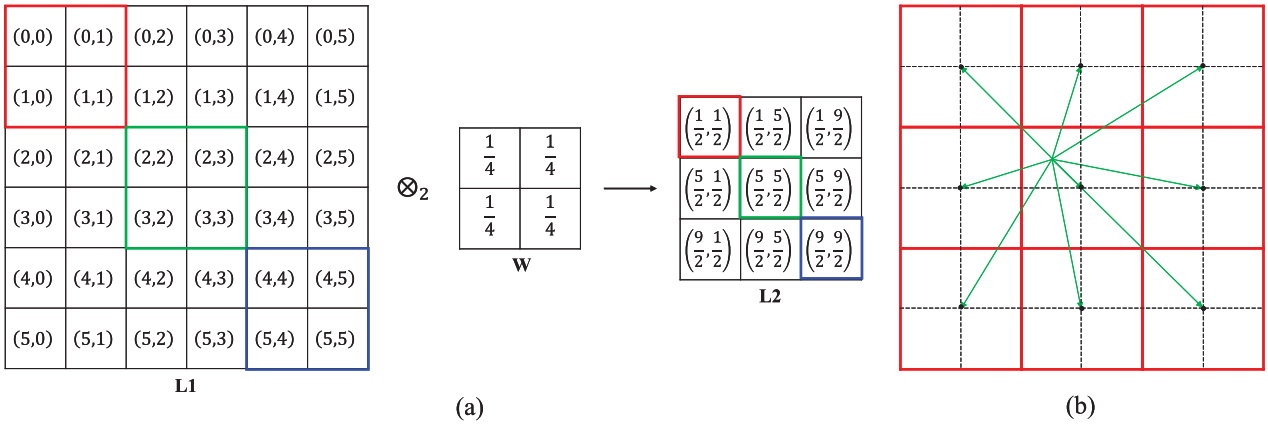


Fig. 5. (a) Convolutional operation on coordinate matrix $\boldsymbol{L1}$, where $\boldsymbol{L1}$ is set to a 6×6 coordinate matrix and the stride R is set to 2. (b) Process of calculating the distance between the coordinates of a token in $\boldsymbol{L1}$ and the center coordinates of all regions in $\boldsymbol{L2}$. In (b), we denote tokens in feature map $\boldsymbol{M}$ with dashed boxes, and all regions in $\boldsymbol{L2}$ with red solid boxes.

as the reduction ratio and merge $R \times R$ tokens into a new region, so that the original N tokens become (N/R^2) regions. The coordinates of the center of each region are used as its coordinates. Therefore, the 2-D positional attention is measured by calculating the distance of N tokens to the center of these (N/R^2) regions. Specifically, the coordinates of these regions are obtained via the convolution on $\boldsymbol{L1}$. We use the convolution kernel $\boldsymbol{W}$ of size $R \times R$ and stride R and set all elements of $\boldsymbol{W}$ to $(1/R^2)$. The formula of this process is denoted as follows:

$$\boldsymbol{L2} = \boldsymbol{L1} \otimes_R \boldsymbol{W} \tag{9}$$

where $\otimes_R$ is the convolution operator with stride R. An example of the convolution operation on $\boldsymbol{L1}$ is shown in Fig. 5(a). Furthermore, Fig. 5(b) shows the process of calculating the distance between a token and all divided regions.

Then we reshape the dimensions of $\boldsymbol{L1}$ and $\boldsymbol{L2}$ to $N \times 2$ and $(N/R^2) \times 2$, respectively, as follows:

$$\boldsymbol{L1}^{(\sqrt{N}\times\sqrt{N}\times 2)} \xrightarrow{\text{Reshape}} \boldsymbol{L1}'^{(N\times 2)} \tag{10}$$

$$\boldsymbol{L2}^{\left(\frac{\sqrt{N}}{R}\times\frac{\sqrt{N}}{R}\times 2\right)} \xrightarrow{\text{Reshape}} \boldsymbol{L2}'^{\left(\frac{N}{R^2}\times 2\right)}. \tag{11}$$

The Euclidean distance between two 2-D coordinates is calculated using (12). Based on it, we use the matrix operation to calculate the Euclidean distance between the two matrices $\boldsymbol{L1}'$ and $\boldsymbol{L2}'$ and get the distance matrix $\boldsymbol{D}$ as shown in (13). Specifically, $\boldsymbol{L1}'$ and $\boldsymbol{L2}'$ include N and (N/R^2) 2-D coordinates, respectively, so the dimension of $\boldsymbol{D}$ is $N \times (N/R^2)$. Hence, the computational complexity of this process is reduced from $O(N^2)$ to $O(N^2/R^2)$

$$d_{\text{Euclidean}}(p_i, p_j) = \sqrt{(x_p - x_q)^2 + (y_p - y_q)^2} \tag{12}$$

$$\begin{aligned} \boldsymbol{D} &= \text{2-D_Dt}(\boldsymbol{L1}', \boldsymbol{L2}') \\ &= \frac{d_{\text{Euclidean}}(\boldsymbol{L1}', \boldsymbol{L2}')}{2R} \end{aligned} \tag{13}$$

$$\text{Gaussian}(x, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{14}$$

$$\begin{aligned} \text{PA} &= \text{2-D_At}(\boldsymbol{L1}', \boldsymbol{L2}') \\ &= \text{Softmax}(\text{Gaussian}(\boldsymbol{D}, \sigma))\boldsymbol{V} \end{aligned} \tag{15}$$

$$\text{SFA} = (1 - \lambda) * \text{SA} + \lambda * \text{PA}. \tag{16}$$

In addition, a token should pay more attention to tokens that are close to it. Therefore, we use a Gaussian function [see (14)] with a mean of 0 and a standard deviation of σ to map relative distances in $\boldsymbol{D}$ to attention values. The 2-D positional attention is calculated as (15). Then the final self-attention is the weighted sum of semantic attention and 2-D positional attention as (16).

The shallow stages of the encoder contain more details, while the deep stages contain richer semantic information. As a result, we vary the stride R and the standard deviation σ of the Gaussian function to adjust the effective range of 2-D positional attention. Specifically, R and σ of the ith stage are set to $\{8, 4, 2, 1\}$ and $\{0.5, 0.5, 0.4, 0.4\}$ for $i = 1, 2, 3, 4$, respectively. As shown in Fig. 6(b), by varying the value of σ, the effective range of 2-D positional attention can be changed. Moreover, we visualize the calculation results of 2-D positional attention in Fig. 6(a) when σ is set to 0.75. The Euclidean distance between the yellow block and the region it belongs to is the smallest, which means that the 2-D positional attention is the largest. As the distance to other regions increases, the 2-D positional attention gradually becomes smaller. Therefore, by adding 2-D positional attention to self-attention, 2-D relative position information can be well preserved.

D. Dilated Residual Connection

In most encoder–decoder-based segmentation methods, feature maps extracted from the input image will be down-sampled multiple times. However, this will result in the loss of details in the image, which hurts the segmentation of small objects and boundaries. Therefore, a skip connection is often used to connect the encoder and the decoder to alleviate this problem. It is effective to use a skip connection to pass details of the shallow feature maps.

However, deep feature maps generated by stages 3 and 4 in the encoder contain limited details and rich global semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

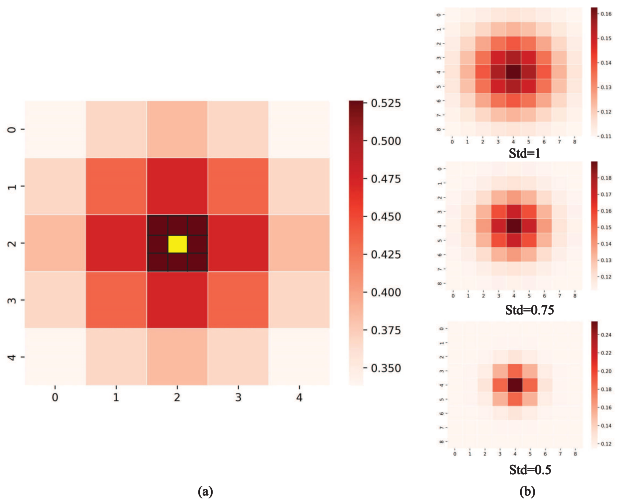


Fig. 6. (a) Visualization of 2-D positional attention and (b) comparison of the effective ranges of 2-D positional attention at different standard deviations. We set $R = 3$ and the dimension of the feature map $\boldsymbol{M}$ is $15 \times 15 \times C$ and $27 \times 27 \times C$ in (a) and (b), respectively. (a) Two-dimensional positional attention from the yellow patch in the center of the feature map $\boldsymbol{M}$ to all 25 regions of size 3×3.

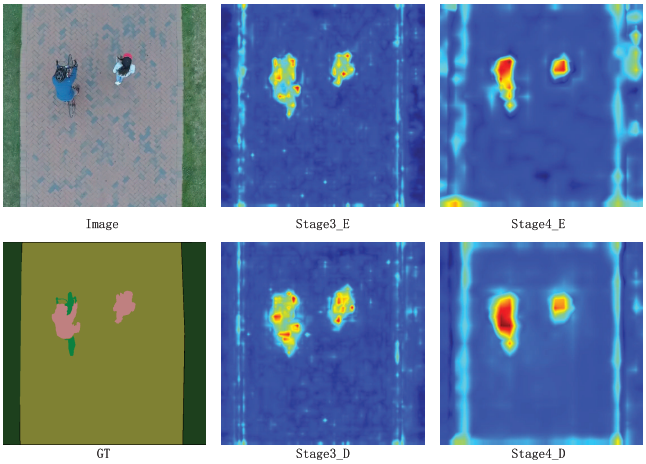


Fig. 7. Heatmaps of the inputs and outputs of dilated residual connections in stages 3 and 4. Stage3_E and Stage4_E represent inputs, and Stage3_D and Stage4_D represent outputs. It can be found that Stage4_E and Stage4_D contain richer global semantic information than Stage3_E and Stage3_D. Moreover, heatmaps of Stage3_D and Stage4_D are smoother and have fewer glitches than heatmaps of Stage3_E and Stage4_E, which benefits from the enlargement of the receptive field brought by the dilated residual connection. GT represents ground truth.

information. It is more necessary to expand the receptive field of deep feature maps to capture global context information better. On this basis, we design the dilated residual connection to further increase the receptive field of deep feature maps. Instead of the skip connection, we use the dilated residual connection to connect the encoder and decoder in stages 3 and 4. While in stages 1 and 2, we still use the skip connection to retain more details for the decoder (see Fig. 2).

The dilated residual connection includes two dilated convolutional layers and a residual connection between the input and the output. The specific structure of the dilated convolution module is shown in Fig. 2, and each convolutional layer is followed by a BatchNorm layer [41] and a ReLU activation function. An example of visualization of the input and output feature maps of the dilated residual connection in stages 3 and 4 is shown in Fig. 7. It is found that the dilated residual connection can further expand the receptive field to better capture the global interactions and the feature maps after the dilated residual connection has richer semantic information.

IV. Experiments

A. Datasets

1) AeroScapes: The AeroScapes aerial semantic segmentation benchmark [42] comprises images captured using a commercial drone from an altitude range of 5–50 m. The dataset provides 3269 (2621 images are used for training, and 648 are used as a validation set for testing) 1280×720 images and ground-truth masks for 12 categories including background.

2) UAVid: The UAVid dataset [43] is a fine-resolution unmanned aerial vehicle (UAV) video dataset for semantic segmentation tasks focusing on urban scenes. UAVid is a challenging benchmark because it has 4 K high-resolution images and complex street scenes with large-scale changes of objects. Specifically, the UAVid dataset includes a total of 420 images, of which 200 are used for training, 70 are used for validation, and 150 are used for testing. The UAVid dataset provides both images and labels for the training and validation sets, but images only for the testing set. Therefore, we adopt the method of the online evaluation, and the evaluation server and the benchmark table are held on Codalab platform.[1]

3) Vaihingen: The International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen dataset[2] contains 33 images in different resolutions and the ground sampling distance (GSD) is 9 cm. We utilize ID: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 32, 34, 37 for training, 30 for validation, and the remaining 17 images for testing. It is worth noting that we only use the near-infrared, red, and green channels in our experiments.

B. Implementation Details

To evaluate the performance of 2DSegFormer comprehensively, we consider some state-of-the-art segmentation methods. We use the mmsegmentation v0.13 [44] codebase and train on a server with two NVIDIA GeForce RTX 3090s. We train the models using an AdamW optimizer with a learning rate (LR) of 0.00006 and a weight decay value of 0.01 for 80, 80, and 40 K iterations on AeroScapes, UAVid, and Vaihingen, respectively. For the LR scheduler, we adopt a poly LR schedule with factor 1 by default. We set batch sizes as 16, 8, and 8 on AeroScapes, Vaihingen, and UAVid, respectively. We crop the raw images as well as corresponding labels into 512×512, 768×768, and 512×512 tokens for AeroScapes, UAVid, and Vaihingen, respectively. Moreover, we augment input images via resizing by a random

[1]https://competitions.codalab.org/competitions/25224
[2]http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE AEROSCAPES DATASET. THE SYMBOL "-" INDICATES UNKNOWN AND BOLD INDICATES THE BEST

Method	Pos.inf.	Backg.	Person	Bike	Car	Drone	Boat	Animal	Obs.	Cons.	Veg.	Road	Sky	mIoU	mF1	OA
FCN [6]	-	89.43	71.58	50.58	93.07	72.90	88.23	59.45	36.98	82.97	96.97	95.53	93.19	66.88	77.57	93.40
PSPNet [45]	-	90.41	70.68	42.44	93.48	73.55	85.86	57.44	33.64	85.72	96.90	96.12	95.21	66.51	76.79	94.03
DeepLabV3+ [46]	-	90.96	69.11	56.50	**94.79**	75.23	89.13	59.33	36.15	85.52	97.17	96.54	93.86	68.46	78.69	94.28
ViT_UperNet [15]	1D APE	90.83	68.28	17.28	92.78	74.33	85.73	61.88	41.14	87.26	97.19	96.24	97.87	66.37	75.90	94.51
SETR [16]	1D APE	89.07	69.24	9.37	88.34	63.10	88.28	65.83	22.78	85.51	97.06	94.75	98.17	63.31	72.62	93.59
Segmenter [48]	1D APE	89.77	65.02	22.32	90.73	72.79	85.54	63.82	27.34	85.05	97.08	95.62	96.77	64.46	74.32	93.86
SegFormer-B0 [40]	-	90.25	68.29	54.19	92.87	72.82	83.79	66.67	37.47	86.57	97.29	95.38	96.42	67.97	78.50	94.11
SegFormer-B1 [40]	-	91.36	69.30	54.96	93.72	76.89	88.27	69.30	38.53	88.30	97.38	96.53	96.86	70.22	80.12	94.81
SegFormer-B2 [40]	-	**91.56**	72.00	**68.27**	93.04	**82.49**	89.12	**73.65**	42.98	88.81	**97.45**	**96.67**	**98.26**	73.40	82.86	**95.05**
SegFormer-B0 + iRPE [21]	2D RPE	89.40	66.01	50.84	91.54	71.41	85.11	53.85	33.68	85.21	97.07	95.40	97.33	65.82	76.40	93.73
2DSegFormer-B0 (ours)	2D PA	90.88	**75.59**	62.18	94.14	73.88	**89.55**	63.88	**49.76**	86.87	97.23	96.12	96.30	71.22	81.37	94.47
2DSegFormer-B1 (ours)	2D PA	91.18	72.68	55.56	94.69	75.48	89.10	69.49	38.32	88.03	97.35	96.47	97.67	70.77	80.50	94.78
2DSegFormer-B2 (ours)	2D PA	91.51	74.91	68.03	94.17	80.39	88.06	72.04	46.84	**88.99**	97.39	96.48	97.61	**73.43**	**83.04**	94.94

scale (from 0.5 to 2), flipping by the horizontal axis with a probability index of 0.5 and randomly rotating an angle (from 45 to 45) with a probability index of 0.5 during training. The cross-entropy loss function is used to measure the difference between the predicted segmentation maps and the ground truth. Also, we use multiscale testing in the inference phase.

C. Evaluation Metrics

We use the overall accuracy (OA), mean intersection over union (mIoU), and mean $F1$ score (mF1) to evaluate the performance. The calculation formulas of OA, mIoU, and mF1 are denoted as follows:

$$\mathrm{OA} = \frac{\sum_{n=1}^{Z} \mathrm{TP}_n}{\sum_{n=1}^{Z} \mathrm{TP}_n + \mathrm{FP}_n + \mathrm{TN}_n + \mathrm{FN}_n} \tag{17}$$

$$\mathrm{mIoU} = \frac{1}{Z} \sum_{n=1}^{Z} \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FP}_n + \mathrm{FN}_n} \tag{18}$$

$$\mathrm{precision} = \frac{1}{Z} \sum_{n=1}^{Z} \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FP}_n} \tag{19}$$

$$\mathrm{recall} = \frac{1}{Z} \sum_{n=1}^{Z} \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FN}_n} \tag{20}$$

$$\mathrm{mF1} = \frac{2}{Z} \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{21}$$

where TP_n, FP_n, TN_n, and FN_n represent the true positive, false positive, true negative, and false negatives, respectively. n and Z represent the class to which a specific object belongs and the number of all interested classes, respectively.

D. Experimental Results

We choose the state-of-the-art CNN-based and transformer-based segmentation methods to compare. To further illustrate the effectiveness of our proposed 2-D positional attention, we introduce iRPE [21], an excellent 2-D position encoding, into SegFormer for experiments. It should be noted that we remove the sequence reduction process in efficient self-attention of MiT to introduce iRPE into SegFormer.

1) Results on the AeroScapes and UAVid Datasets: First, we verify and test the effectiveness of 2DSegFormer by comparing it with the state-of-the-art methods on two commonly used aerial image datasets. Table I shows the comparison with the state-of-the-art methods on the AeroScapes dataset. From Table I, it is found that 2DSegFormer achieves better performance on the AeroScapes dataset. Moreover, 2DSegFormer has a better performance than SegFormer-B0, B1, B2, especially B0. Compared with other position encoding methods, 2-D positional attention can bring greater benefits, especially for the segmentation of small-size objects such as the three classes of person, obstacle, and bike. This shows that 2-D positional attention better records the 2-D positional relationships of tokens. Examples of semantic segmentation results on the AeroScapes dataset are shown in Fig. 8. By comparing the visualization results, we find that 2DSegFormer is better than other excellent methods in the segmentation of details. This is due to the local context retained through 2-D positional attention and the global semantic information captured through the dilated residual connections.

Because the labels of the UAVid test set are not public, we submit the prediction results of the test set to its official website for verification and evaluation. The experimental results of our proposed 2DSegFormer and the state-of-the-art methods on the UAVid test set are listed in Table II. Moreover, we visually compare the prediction results of several representative methods on the UAVid test set in Fig. 9. From the visualization results, it is found that our 2DSegFormer is better at the segmentation of details and easily confusing objects than other methods. This is because 2DSegFormer has richer spatial details and global semantic context information. It is worth mentioning that our best model 2DSegFormer-B2 achieves first place in the public ranking (72.68% mIoU) of the UAVid official evaluation server.

2) Results on the Vaihingen Dataset: To verify that 2DSegFormer is also robust in the semantic segmentation on remote-sensing images, we do experiments on the ISPRS Vaihingen dataset which is commonly used for remote-sensing semantic segmentation. By the comparison in Table III, 2DSegFormer also achieves competitive results on the ISPRS Vaihingen

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

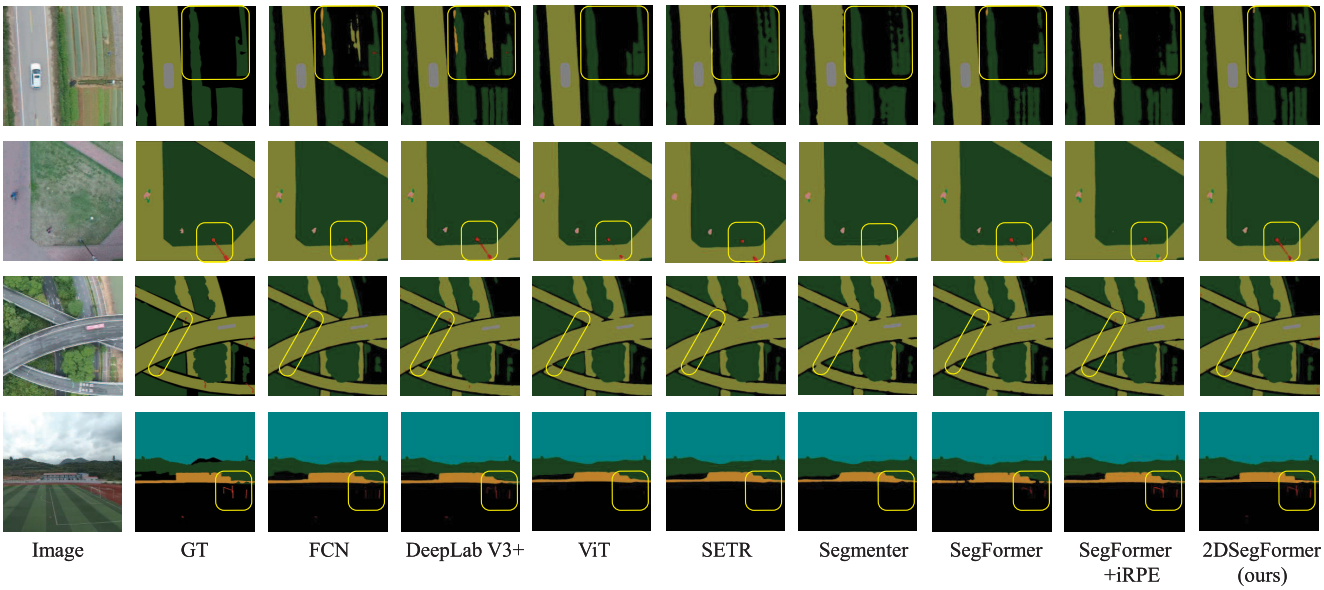


Fig. 8. Examples of semantic segmentation results on the AeroScapes dataset. GT represents ground truth. Several obvious differences are marked with yellow boxes.

TABLE II

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE UAVID TEST SET. THE SEGMENTATION RESULTS OF THE FIRST SIX METHODS ARE CITED FROM [39]. THE SYMBOL "-" INDICATES UNKNOWN AND BOLD INDICATES THE BEST

Method	Pos.inf.	Clutter	Building	Road	Tree	Lowveg.	MovingCar	StaticCar	Human	mIoU	OA
MSD [43]	-	57.0	79.8	74.0	74.5	55.9	62.9	32.1	19.7	57.0	-
MANet [29]	-	64.5	85.4	77.8	77.0	60.3	67.2	53.6	14.9	62.6	-
ABCNet [28]	-	67.4	86.4	81.2	79.9	63.1	69.8	48.4	13.9	63.8	-
BANet [38]	-	66.7	85.4	80.7	78.9	62.1	69.3	52.8	21.0	64.6	-
BoTNet [47]	2D RPE	64.5	84.9	78.6	77.4	60.5	65.8	51.9	22.4	63.2	-
UNetFormer [39]	1D RPE	68.4	87.4	81.5	80.2	63.5	73.6	56.4	31.0	67.8	-
ViT_Upernet [15]	1D APE	71.72	89.97	**83.42**	81.63	66.45	75.69	60.54	23.76	69.15	88.24
SETR [16]	1D APE	70.99	89.45	82.50	81.33	65.84	75.37	64.76	21.91	69.02	87.92
Segmenter [48]	1D APE	70.95	89.66	82.55	81.64	66.31	74.38	62.81	20.73	68.63	88.01
SegFormer-B0 [40]	-	69.87	88.90	81.60	81.68	66.52	74.87	62.79	26.23	69.06	87.68
SegFormer-B1 [40]	-	71.02	89.45	82.25	81.71	66.52	75.97	70.96	28.58	70.81	88.06
SegFormer-B2 [40]	-	**72.41**	**90.46**	83.38	82.11	66.85	77.39	75.91	31.32	72.48	88.65
SegFormer-B0 + iRPE [21]	2D RPE	66.66	86.75	79.35	79.76	63.82	72.33	60.43	22.48	66.45	86.13
2DSegFormer-B0 (ours)	2D PA	70.36	89.20	81.93	81.68	66.54	76.51	69.66	28.76	70.58	87.91
2DSegFormer-B1 (ours)	2D PA	71.62	89.79	82.89	81.93	66.91	76.63	69.61	29.90	71.16	88.31
2DSegFormer-B2 (ours)	2D PA	72.39	90.45	83.30	**82.20**	**66.93**	**78.20**	**76.40**	**31.60**	**72.68**	**88.66**

dataset. In particular, 2DSegFormer improves the segmentation accuracy of small-size objects such as cars, which is due to the introduction of 2-D positional attention. Because 2-D positional attention provides enough 2-D position information for the model to obtain rich local spatial context, which is crucial for the segmentation of small-size objects in aerial images. It should be noted that we use ground reference data without eroded boundaries when we do experiments on the Vaihingen dataset, so the evaluation indices are not as high as reported in certain elements of the literature.

E. Ablation Studies

In the 2DSegFormer, 2-D positional attention is used to help the model better learn the positional relationships of tokens. Furthermore, the dilated residual connection is used to expand the receptive field of the deep feature maps to achieve fine segmentation. To further evaluate the effectiveness of our proposed 2-D positional attention and dilated residual connection, we do a lot of ablation experiments on the AeroScapes validation set.

1) Encoder and Decoder: We use Segformer-B0 as the baseline and verify the performance of our proposed 2-D positional attention and dilated convolution module by replacing its encoder and decoder. It can be found in Table IV that using the encoder and decoder, we proposed an improvement of 3.25 on the mIoU. In addition, when the encoder or the decoder is fixed, choosing our designed decoder or encoder can achieve better performance, except when the decoder is fixed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

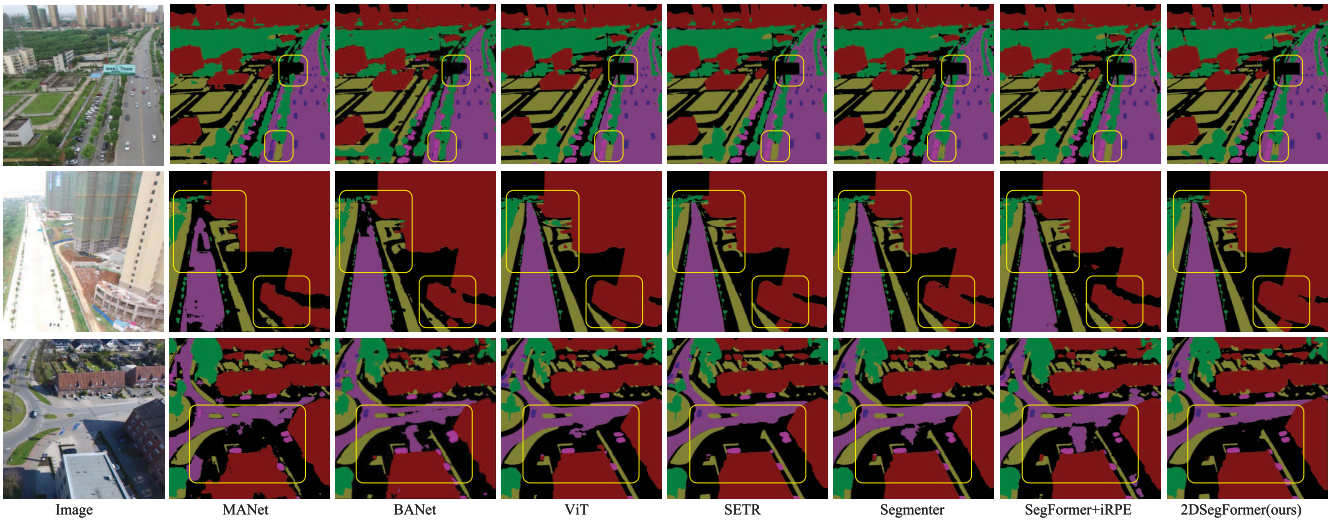


Fig. 9. Examples of semantic segmentation results on the UAVid test set, where the prediction results of MANet and BANet are downloaded from the official evaluation platform. Several obvious differences are marked with yellow boxes.

TABLE III

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE VAIHINGEN DATASET. THE SEGMENTATION RESULTS OF THE FIRST FOUR METHODS ARE CITED FROM [29]. THE SYMBOL "-" INDICATES UNKNOWN AND BOLD INDICATES THE BEST

Method	Pos.inf.	Imp.surf.	Buildings	Lowveg.	Trees	Cars	mIoU	mF1	OA
U-Net [12]	-	-	-	-	-	-	68.68	81.03	84.03
DeeplabV3+ [46]	-	-	-	-	-	-	72.30	83.58	85.93
PSPNet [45]	-	-	-	-	-	-	72.45	83.64	86.37
MANet [29]	-	-	-	-	-	-	74.92	85.42	86.96
Segmenter [48]	1D APE	87.74	91.59	77.39	85.51	54.97	67.67	79.44	77.42
ViT [15]	1D APE	89.42	93.41	80.35	86.69	61.36	71.28	82.25	87.59
SETR [16]	1D APE	89.38	93.23	79.79	86.34	67.17	72.21	83.18	87.39
SegFormer-B0 [40]	-	90.12	93.72	81.06	86.89	76.82	75.51	85.72	88.19
SegFormer-B1 [40]	-	90.64	94.10	81.21	87.16	78.32	76.35	86.29	88.57
SegFormer-B2 [40]	-	90.75	94.39	81.30	87.15	80.14	77.00	86.74	88.71
SegFormer-B0 + iRPE [21]	2D RPE	89.68	93.47	80.05	86.65	72.71	73.87	84.51	87.74
2DSegFormer-B0 (ours)	2D PA	90.35	94.09	81.01	86.94	79.46	76.42	86.37	88.38
2DSegFormer-B1 (ours)	2D PA	90.60	94.35	81.40	87.12	80.08	76.94	86.71	88.67
2DSegFormer-B2 (ours)	2D PA	**90.96**	**94.50**	**81.44**	**87.20**	**81.29**	**77.49**	**87.08**	**88.85**

TABLE IV

ABLATION EXPERIMENT OF ENCODER (MYT-B0) AND DECODER (UHEAD) IN 2DSEGFORMER-B0 ON THE AEROSCAPES DATASET, WHERE MIT-B0 AND SEGFORMERHEAD ARE THE ENCODER AND DECODER OF SEGFORMER-B0, RESPECTIVELY

Method	Encoder	Decoder	mIoU	OA	mF1
SegFormer-B0	MiT-B0	SegFormerHead	67.97	94.11	78.50
2DSegFormer-B0	MiT-B0	UHead (ours)	70.75	94.40	80.87
	MyT-B0 (ours)	SegFormerHead	67.67	94.03	78.27
	MyT-B0 (ours)	UHead (ours)	**71.22**	**94.47**	**81.37**

as SegFormerHead. The experimental results further verify the effectiveness of our proposed 2-D positional attention and the decoder including the dilated residual connection.

2) 2-D Positional Attention: Then we add 2-D positional attention into MiT at different stages of the encoder to get a more powerful encoder. Specifically, the ablation experiment results of using 2-D positional attention at different stages are shown in Table V and the checkmark means that 2-D positional attention is added to the self-attention of the corresponding stage. It can be found that when the 2-D positional attention is added to the self-attention of all four stages, the performance is the best. This also proves again the importance of 2-D position information for ViTs. In addition, because the shallow feature maps contain more details, the relative positional relationships between tokens are particularly important in the shallow layers. Therefore, just adding 2-D positional attention to stage 1 can already get great benefits. By ablation results and the above analysis, it is effective to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION EXPERIMENT OF USING 2-D POSITIONAL ATTENTION AT DIFFERENT STAGES OF 2DSEGFORMER-B0 ON THE AEROSCAPES DATASET

Method	Stage1	Stage2	Stage3	Stage4	mIoU	OA	mF1
SegFormer-B0	-	-	-	-	67.97	94.11	78.50
2DSegFormer-B0	√				70.11	94.38	80.16
	√	√			69.29	94.14	79.81
	√	√	√		70.96	**94.58**	81.11
	√	√	√	√	**71.22**	94.47	**81.37**

TABLE VI
ABLATION EXPERIMENT OF THE STANDARD DEVIATION IN GAUSSIAN FUNCTION ON THE AEROSCAPES DATASET

Method	Standard deviation	mIoU	OA	mF1
SegFormer-B0	-	67.97	94.11	78.50
2DSegFormer-B0	{0.5, 0.5, 0.5, 0.5}	69.34	94.43	79.40
	{0.4, 0.4, 0.4, 0.4}	69.64	93.70	80.13
	{0.5, 0.5, 0.4, 0.4}	**71.22**	**94.47**	**81.37**

TABLE VII
ABLATION EXPERIMENT OF USING DILATED RESIDUAL CONNECTION AT DIFFERENT STAGES OF 2DSEGFORMER-B0 ON THE AEROSCAPES DATASET

Method	Stage1	Stage2	Stage3	Stage4	mIoU	OA	mF1
SegFormer-B0	-	-	-	-	67.97	94.11	78.50
2DSegFormer-B0	√	√	√	√	70.19	94.22	80.10
		√	√	√	69.60	94.33	79.85
			√	√	**71.22**	94.47	**81.37**
				√	70.79	**94.56**	80.90

TABLE VIII
ABLATION EXPERIMENT OF THE DILATION RATES IN DILATED RESIDUAL CONNECTION ON THE AEROSCAPES DATASET

Method	Dilation rate	mIoU	OA	mF1
SegFormer-B0	-	67.97	94.11	78.50
2DSegFormer-B0	{2, 2}	69.65	94.07	79.97
	{3, 3}	69.69	94.14	80.02
	{2, 3}	**71.22**	**94.47**	**81.37**
	{2, 4}	70.27	94.00	80.48

record 2-D position information for ViTs using our proposed 2-D positional attention.

We also do an ablation experiment of the standard deviation in the Gaussian function. It can be seen from Table VI that when the standard deviations in the four stages are the same, 0.4 is the better choice. When the standard deviations in the four stages are set to {0.5, 0.5, 0.4, 0.4}, respectively, better performance can be obtained. We believe that the reason for this phenomenon is that the feature maps in stages 1 and 2 contain more spatial information, so a larger effective range of 2-D positional attention is required. However, the feature maps in the latter two stages contain richer global semantic information, so the effective range should be reduced.

3) Dilated Residual Connection: Since feature maps of different sizes are obtained, we can use the dilated residual connection to replace the skip connection in the encoder–decoder structure at these four stages. Therefore, we do ablation experiments using the dilated residual connection at different stages. It can be found from Table VII that better performance can be achieved when using the dilated residual connection in stages 3 and 4 and skip connections in stages 1 and 2. Because the shallow feature maps contain richer details, using a skip connection to pass details directly to the corresponding stages of the decoder can alleviate the loss of details caused by down-sampling. Instead, the deep feature maps have more global semantic information. To capture more global semantic context information, using the dilated residual connection to pass details and enlarge the receptive field further is better.

In the dilated residual connection, we use two convolutional layers with dilation rates of 2 and 3, respectively. Therefore, we also do ablation experiments on the dilatation rates of these two convolutional layers in the dilated residual connection. In ablation experiments, we set dilatation rates to {2, 2}, {2, 3}, {3, 3}, and {2, 4}, respectively. It is found from Table VIII that when the dilatation rate is set to {2, 3}, 2DSegFormer achieves better performance.

TABLE IX
COMPARISON OF COMPUTATIONAL COMPLEXITY AND MODEL PARAMETERS. THE SMALLER THE VALUE OF FLOPS, THE FASTER THE INFERENCE SPEED OF THE MODEL

Method	Input Shape	FLOPs (G)	Params (M)
FCN [6]	3×256×256	13.87	12.82
DeeplabV3+ [46]		13.56	12.47
PSPNet [45]		13.57	12.79
ViT_UperNet [15]		8.73	35.00
SETR [16]		2.66	9.93
Segmenter [48]		1.13	6.69
SegFormer-B0 [40]		1.76	3.72
SegFormer-B1 [40]		3.59	13.74
SegFormer-B2 [40]		14.44	27.36
SegFormer-B0 + iRPE [21]		4.18	3.76
2DSegFormer-B0 (ours)		5.04	7.51
2DSegFormer-B1 (ours)		7.37	22.94
2DSegFormer-B2 (ours)		42.93	52.35

F. Efficiency Analysis

For comprehensive comparisons, Table IX lists the calculations and parameters of the state-of-the-art models in the same environment. Specifically, we use floating point operations (FLOPs) to measure the computational complexity of the model, and the smaller the value, the faster the calculation speed. Meanwhile, the number of parameters of each model is quantitatively compared. Compared with CNN-based models, our 2DSegForemr-B0 not only has advantages in both parameters and computational complexity, but also achieves better performance. Moreover, we get a considerable performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

improvement compared to SegForemr-B0 by adding a few parameters and calculation costs. 2DSegFormer-B2 has a larger number of parameters and a longer inference time. The increase in the number of parameters and inference time is mainly due to dilated residual connections. In future work, we will also consider the possibility of further optimizing 2DSegFormer-B2 to reduce the network parameters.

V. Conclusion

In this article, we propose the 2DSegFormer for semantic segmentation on aerial images. Specifically, we design a 2-D positional attention mechanism to help ViTs retain the relative 2-D position information between tokens. Two-dimensional positional attention can help the transformer-based model better understand the local context information to segment the details and small-size objects. In addition, we propose the dilated residual connection to replace skip connections in stages 3 and 4 to help expand the effective receptive field of deep feature maps. Through a large number of experiments on the three commonly used aerial and remote sensing image datasets, the performance of our 2DSegFormer is superior to the state-of-the-art methods. After sufficient ablation experiments and comparisons, the effectiveness of our proposed 2-D positional attention and the dilated residual connection has been verified. Moreover, the segmentation accuracy has been greatly improved with a small increase in parameters. In the future, we will try to add 2-D positional attention to more ViTs and explore more possibilities for its fusion with the self-attention mechanism.

References

[1] W. Sun, K. Ren, X. Meng, C. Xiao, G. Yang, and J. Peng, "A band divide-and-conquer multispectral and hyperspectral image fusion method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[2] W. Sun et al., "MLR-DBPFN: A multi-scale low rank deep back projection fusion network for anti-noise hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522914.

[3] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[4] Y. Feng, X. Sun, W. Diao, J. Li, X. Gao, and K. Fu, "Continual learning with structured inheritance for semantic segmentation in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[5] S. M. Azimi, P. Fischer, M. Korner, and P. Reinartz, "Aerial LaneNet: Lane-marking semantic segmentation in aerial imagery using wavelet-enhanced cost-sensitive symmetric fully convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2920–2938, May 2019.

[6] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 3431–3440.

[7] Z. Chen et al., "Vehicle detection in high-resolution aerial images via sparse representation and superpixels," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 103–116, Jan. 2016.

[8] H. Costa, G. M. Foody, and D. S. Boyd, "Supervised methods of image segmentation accuracy assessment in land cover mapping," *Remote Sens. Environ.*, vol. 205, pp. 338–351, Feb. 2018.

[9] Z. Guo et al., "Semantic segmentation for urban planning maps based on U-Net," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 6187–6190.

[10] J. Zhang, Q. Hu, J. Li, and M. Ai, "Learning from GPS trajectories of floating car for CNN-based urban road extraction with high-resolution satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1836–1847, Mar. 2021.

[11] L. Ding and L. Bruzzone, "DiResNet: Direction-aware residual network for road extraction in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10243–10254, Dec. 2021.

[12] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[13] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2015.

[14] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[15] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[16] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 6881–6890.

[17] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[18] J. Gehring, M. Auli, D. Grangier, D. Yarats, and Y. N. Dauphin, "Convolutional sequence to sequence learning," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1243–1252.

[19] P. Shaw, J. Uszkoreit, and A. Vaswani, "Self-attention with relative position representations," 2018, *arXiv:1803.02155*.

[20] P. Ramachandran, N. Parmar, A. Vaswani, I. Bello, A. Levskaya, and J. Shlens, "Stand-alone self-attention in vision models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–13.

[21] K. Wu, H. Peng, M. Chen, J. Fu, and H. Chao, "Rethinking and improving relative position encoding for vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10033–10041.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[23] W. Zhao and S. Du, "Learning multiscale and deep representations for classifying remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 113, pp. 155–165, Mar. 2016.

[24] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[25] L. Wang, C. Zhang, R. Li, C. Duan, X. Meng, and P. M. Atkinson, "Scale-aware neural network for semantic segmentation of multi-resolution remote sensing images," *Remote Sens.*, vol. 13, no. 24, p. 5015, Dec. 2021.

[26] Z. Li et al., "Cascaded multiscale structure with self-smoothing atrous convolution for semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5605713.

[27] D. He, Q. Shi, X. Liu, Y. Zhong, and L. Zhang, "Generating 2m fine-scale urban tree cover product over 34 metropolises in China based on deep context-aware sub-pixel mapping network," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 106, Feb. 2022, Art. no. 102667.

[28] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[29] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *EEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607713.

[30] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[31] Z. Chen, D. Li, W. Fan, H. Guan, C. Wang, and J. Li, "Self-attention in reconstruction bias U-Net for semantic segmentation of building rooftops in optical remote sensing images," *Remote Sens.*, vol. 13, no. 13, p. 2524, Jun. 2021.

[32] Z. Huang, Q. Zhang, and G. Zhang, "MLCRNet: Multi-level context refinement for semantic segmentation in aerial images," *Remote Sens.*, vol. 14, no. 6, p. 1498, Mar. 2022.

[33] D. He, Q. Shi, X. Liu, and Y. Zhong, "Deep subpixel mapping based on semantic information modulated network for urban land use mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10628–10646, Dec. 2021.

[34] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.

[35] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 10012–10022.
[36] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.
[37] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.
[38] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021.
[39] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, 2022.
[40] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021.
[41] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. 32nd Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
[42] I. Nigam, C. Huang, and D. Ramanan, "Ensemble knowledge transfer for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1499–1508.
[43] Y. Lyu, G. Vosselman, G.-S. Xia, A. Yilmaz, and M. Y. Yang, "UAVid: A semantic segmentation dataset for UAV imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 108–119, Jul. 2020.
[44] M. Contributors. *MMSegmentation: Openmmlab Semantic Segmentation Toolbox and Benchmark*. Accessed: May 24, 2021. [Online]. Available: https://github.com/open-mmlab/mmsegmentation
[45] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[46] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[47] A. Srinivas, T.-Y. Lin, N. Parmar, J. Shlens, P. Abbeel, and A. Vaswani, "Bottleneck transformers for visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16519–16529.
[48] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.



Xinyu Li received the B.Sc. degree in communication engineering from the Hunan Institute of Science and Technology, Yueyang, China, in 2020. He is currently pursuing the M.Sc. degree with the Department of Communication Engineering, Guangdong University of Technology, Guangzhou, China.

His primary research interests include semantic segmentation and vision transformer.



Yu Cheng (Member, IEEE) received the bachelor's and master's degrees in computer science from Xiangtan University, Xiangtan, China, in 2000 and 2005, respectively, and the Ph.D. degree in computer science from Zhejiang University, Hangzhou, China, in 2008.

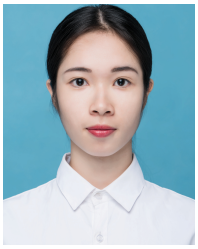
His research interests include the fields of deep learning, semantic segmentation, and object detection.



Yi Fang (Senior Member, IEEE) received the Ph.D. degree in communication engineering from Xiamen University, Xiamen, China, in 2013.

From May 2012 to July 2012, he was a Research Assistant in electronic and information engineering with The Hong Kong Polytechnic University, Hong Kong. From September 2012 to September 2013, he was a Visiting Scholar in electronic and electrical engineering with University College London, London, U.K. From February 2014 to February 2015, he was a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. He is currently a Full Professor and the Vice Dean with the School of Information Engineering, Guangdong University of Technology, Guangzhou, China. His research interests include information and coding theory, spread-spectrum modulation, and cooperative communications.

Dr. Fang served as the Publicity Co-Chair for the International Symposium on Turbo Codes and Iterative Information Processing 2018.



Hongmei Liang received the B.Sc. degree in communication engineering from Guangxi Normal University, Guilin, China, in 2020. She is currently pursuing the M.Sc. degree with the Department of Communication Engineering, Guangdong University of Technology, Guangzhou, China.

Her primary research interests include computer vision.



Shaoqiu Xu received the Ph.D. degree in electrical engineering and electronics from the University of Liverpool, Liverpool, U.K., in 2004.

She is currently an Associate Professor with the School of Information Engineering, Guangdong University of Technology, Guangzhou, China. Her research interests include computer vision, machine learning, and artificial intelligence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:34:51 UTC from IEEE Xplore. Restrictions apply.